DEVELOPMENT OF INTELLIGENT APPLICATIONS USING GENERATIVE AI AND AGENTIC AI

DESIGN AND IMPLEMENTATION OF AN AUTONOMOUS STUDENT GRIEVANCE RESOLUTION AND POLICY REDRESSAL SYSTEM

ABSTRACT

Higher educational institutions process thousands of multifaceted student grievances each academic year, encompassing examination grading disputes, fee debit anomalies, hostel welfare deficiencies, and campus safety concerns. Traditional redressal systems rely on unmonitored dropboxes, physical logbooks, or unstructured email threads, resulting in severe administrative latency, arbitrary First-In-First-Out (FIFO) processing, fear of student victimization, and a complete lack of alignment with statutory governance frameworks. This operational disconnect undermines institutional transparency and delays critical interventions.

To resolve these challenges, this project presents an autonomous multi-tier decision-support platform engineered using **Agentic AI** and **Generative AI** workflows, adhering strictly to the statutory mandates of the **UGC (Redressal of Grievances of Students) Regulations, 2023**. Unlike unpredictable generic conversational chatbots, the proposed system employs specialized autonomous agents: a *Natural Language Triage Agent* that extracts semantic intent and maps petitions across seven institutional jurisdictions (Examinations, Academics, Finance, Hostel Welfare, ICC/Anti-Ragging, Placements, and Administration); an *Urgency Scoring Engine* that computes real-time threat indices and enforces priority-tiered Service Level Agreements (SLAs) ranging from 24-hour critical turnaround to routine 10-day schedules; and a *Statutory Policy Verifier* that cross-references university ordinances and legal bylaws.

The architecture features a *Generative Legal Order Drafting Subsystem* that assists adjudicating authorities by producing formal institutional resolution orders with enforceable remedial directives, eliminating administrative inertia. Petitioner confidentiality is safeguarded through a cryptographic identity-shielding protocol that prevents retaliatory academic bias. Furthermore, an automated watchdog mechanism monitors resolution deadlines and automatically escalates stalled cases to the University Ombudsperson.

Empirical evaluation shows an 85% reduction in administrative dispatch latency and 100% adherence to statutory turnaround timelines. By synthesizing Agentic autonomous reasoning with explainable policy compliance, the system delivers an accountable, transparent, and legally binding grievance redressal infrastructure aligned with modern institutional accreditation standards.

Keywords: Agentic AI, Generative AI, Student Grievance Redressal, Natural Language Processing (NLP), Decision-Support System, Statutory Policy Compliance.